TECHNICAL TESTING REPORT

ILLUSTRATIVE AND UNVERIFIED — NO ACTUAL TEST EXECUTION. All example outcomes, timings, defects, tester references and environment assumptions are hypothetical. Unknown identities, dates and build information remain placeholders. Actual executions: 0.

Title : GradeSense AI System Testing

Group Name : [Group name]
Technical Evaluators (IT Experts): [IT evaluator names and credentials - unverified]

Instructions :

1)This report is to be done and tested by the programmers and system analysts of the group and 10 IT experts .

2) Fill out the details before starting the testing procedure . LIST ALL THE MODULES / FUNCTIONALITY OF YOUR SYSTEM, DELETE THOSE WHO ARE NOT INCLUDED IN YOUR SYSTEM. (YOU CAN ALSO GET THEM FROM YOUR MIDTERM PROGRESS REPORT IF AVAILABLE)
3. Record and document the testing specially when there are bugs or errors.
4) Functional Testing / Test Case or / Testing Plan (copy paste when you are done to your documentation Testing and Procedures)
5) When the testing is completed , errors are fixed then you can proceed to SYSTEM EVALUATION.

COMPREHENSIVE SOFTWARE TECHNICAL TESTING REPORT

1. Document Information

Item	Details
Project/System Name	GradeSense AI
Project Title	GradeSense AI grade management and AI assessment system
Version	[Tested application version / commit - unverified]
Testing Date	[Actual testing date - not executed]
Testing Environment	Illustrative isolated environment; all Section 6 specifications assumed/unverified
Tested By	[Tester names / roles - unverified]; [10 IT expert names - unverified]
Developer/Development Team	[Development team / institution]
Test Report Version	1.0 illustrative draft
Status	Draft - illustrative/unverified; actual execution and approval pending

2. Executive Summary

2.1 Purpose of Testing

This is an illustrative, unverified system testing report for GradeSense AI. It provides executable scenarios, proposed acceptance criteria and hypothetical outcomes. No system tests, benchmarks, expert evaluations or defect retests were performed for this report. Actual executions recorded: 0. The example set contains 72 cases, 68 illustrative passes and 4 illustrative failures across 3 hypothetical defects; it does not establish deployment readiness.

2.2 Testing Objectives

The planned testing objectives are to:

1. Verify that all major system functions operate according to specifications.
2. Identify defects, errors, and unexpected system behavior.
3. Evaluate system performance under defined conditions.
4. Verify compatibility across supported devices, browsers, and operating environments.
5. Assess system security and access-control mechanisms.
6. Verify database operations and data integrity.
7. Evaluate the system against applicable software quality standards.
8. Confirm corrections through actual retesting before closing any verified defect.
9. Establish readiness only after real execution evidence and authorized review.

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3. System Overview

3.1 System Description

System Name: GradeSense AI

Description:

GradeSense AI is a MERN grade-management and assessment application with administrator, teacher and student roles. It supports subjects and enrollment, manual/AI assessment drafts, publication, timed student submissions, objective and rubric-based essay grading, teacher review, analytics, notifications, Quiz Arena, and PDF/Excel reports. The available project files establish feature context only; deployed behavior remains unverified.

3.2 Intended Users

User Type	Description	Access Level
Administrator	Manage accounts, dashboards and audit/monitoring features.	Admin-authorized records; backup/restore is mock only.
Faculty/Staff (Teacher)	Manage owned subjects, assessments, submissions and reviewed scores.	Owned teacher records and enrolled learners.
Student/User	Take assigned assessments; review own grades and feedback.	Enrolled published/closed assessments and own records.

3.3 Major System Modules

No.	Module/Feature	Description
1	User Authentication	Registration, login/logout, JWT and role guards.
2	Dashboard	Teacher, student and admin summary views.

No.	Module/Feature	Description
3	User Management	Admin account management and role visibility.
4	Subjects and Enrollment	Teacher-owned subjects and assigned learners.
5	Assessment Lifecycle	Draft, edit, review, publish, close and timing.
6	AI Quiz Generation	Lesson upload/text, validation and bounded fallback.
7	Submission and Grading	Objective keys, rubric essays and pending review.
8	Teacher Review	Score overrides, reasons and final grade reconciliation.
9	Reports and Analytics	PDF/Excel exports, history and progress.
10	Notifications and Announcements	Role-appropriate messages and destinations.
11	Quiz Arena	Game participation, attempts and results.
12	Database and Audit	MongoDB persistence, indexes and audit records; admin recovery mocks.

4. Testing Scope

4.1 In-Scope

The following components are covered by the illustrative scenarios:

- User registration and authentication
- Login/logout
- User account management
- Subjects, assessments, submission, grading, teacher overrides and Quiz Arena
- Database operations
- Input validation
- Error handling
- Navigation
- API integration
- Role, ownership, enrollment, answer-key confidentiality and token checks
- Performance
- Browser/device compatibility
- Data storage and retrieval
- Teacher/student grade views, analytics, PDF/Excel reports and notifications

4.2 Out-of-Scope

The following are excluded from the example execution totals:

> Live cloud/provider verification; production migration; destructive production changes; real student records; institution-specific grading policies; legal/compliance certification; full penetration testing; expert/user acceptance ratings; statistical AI fairness or accuracy claims.

Real backup/restore and disaster recovery are unverified: admin backup/restore handlers are mocks. Scanned-PDF OCR, cross-instance generation idempotency, long-duration availability and infrastructure failover need separate evidence. Programming content may be assessed as text; secure execution of arbitrary student code is not assumed.

5. Testing Methodology

Proposed approach: requirements-based scenarios, synthetic fixtures, boundary and negative cases, controlled provider fault injection, database reconciliation and evidence capture. All expected behaviors are proposed acceptance criteria, not proof of implementation.

Requirements → Planning → Case Development → Actual Execution → Defect Verification → Correction → Retesting → Evaluation

Testing Types

Testing Type	Purpose	Performed?
Unit Testing	Existing test files provide design context; no suite run for this report.	Not executed
Integration Testing	Verify score/report and provider/persistence boundaries.	Not executed
System Testing	Exercise full teacher-to-student assessment workflow.	Not executed
Functional Testing	Execute F-001..020 against approved requirements.	Not executed
Usability Testing	Use UI-001..006 and documented user tasks.	Not executed
Performance Testing	Use PT-001..005; all numbers are illustrative.	Not executed
Security Testing	Use SEC-001..006 and privacy checks.	Not executed
Compatibility Testing	Use BR/DEV matrices with exact versions recorded.	Not executed

Testing Type	Purpose	Performed?
Database Testing	Use DB-001..006 and reconciliation checks.	Not executed
API Testing	Use API-001..008 with controlled faults.	Not executed
Regression Testing	Use REG-001..005 after verified changes.	Not executed
User Acceptance Testing	Programmer/analyst and 10-IT-expert evaluation remains pending.	Not executed

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Status legend throughout: ILL-PASS and ILL-FAIL mean illustrative, unverified outcomes only. Example means a fabricated execution narrative. Assumed means an environment or data proposal, not an observed configuration. Actual status for every scenario is NOT EXECUTED. The template requirement for 10 IT experts remains pending; no expert participation is claimed.

6. Test Environment

6.1 Hardware Environment

Component	Specification
Processor	Assumed: client four-core CPU; server 2 vCPU. Actual: [verify].
RAM	Assumed: client 8 GB; API server 4 GB; DB 4 GB. Actual: [verify].
Storage	Assumed: client SSD; server/DB 20 GB available. Actual: [verify].

Component	Specification
Display Resolution	Assumed: desktop 1920×1080; laptop 1366×768; mobile 390×844 CSS px.
Network	Assumed: 20 Mbps client link; 30–80 ms RTT; stable test LAN/cloud route.
Other	Assumed: isolated test host, synthetic records, one API instance; no production changes.

6.2 Software Environment

Software	Version
Operating System	Assumed: Windows 11 client; Linux server [exact release].
Web Browser	Assumed: Chrome/Edge/Firefox/Safari [exact versions pending].
Programming Language	Assumed: JavaScript; Node.js [exact supported version]; no runtime verified.
Framework	Assumed: React/Vite client; Express API; exact installed versions [verify lockfiles].
Database	Assumed: MongoDB [version]; Mongoose [version]; required indexes present [verify].
Web Server	Assumed: Node/Express [version], local API port 5000; proxy [verify].
API/LLM Service	Assumed: Groq primary, OpenAI fallback; models [record actual configured IDs]. Stub responses for deterministic faults.

Software	Version
Other Dependencies	Assumed: JWT, bcryptjs, Multer, Socket.IO, PDFKit, ExcelJS and document parsers [versions pending].

All entries in 6.1 and 6.2 are illustrative environment assumptions. Freeze exact runtime, browser, database, dependency and model versions against [build/commit] before actual execution. Do not use production credentials or records.

7. Test Data

Assumed synthetic fixture only. No fixture was loaded or test record created for this report. Use reserved .test email addresses, artificial names and reviewed lesson text; reset the isolated database between mutation scenarios.

Data Category	Description	Number of Records
User Accounts	Assumed: 1 admin, 2 teachers and 100 students; artificial names and .test emails.	103 proposed
Sample Transactions	Assumed: controlled assessment/submission fixtures; reset per case.	25 assessment summaries; 100 submissions proposed
Test Questions/Content	Assumed: A-grade 12 items/40 pts; A-AI 10 items/36 pts; no real lesson uploads.	22 primary fixture items
Database Records	Assumed: subject A/T1 and B/T2; enrollments and notifications generated only in test DB.	2 subjects; dependent records [fixture-specific]

Data Category	Description	Number of Records
Other	Assumed: 4 readable file types, scanned control, malformed/empty/oversize controls and provider stubs.	4 valid files; 5 invalid/boundary files proposed

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Fixture arithmetic: A-grade has ten two-point objective items and two ten-point essays, maximum 40. A proposed run yields 18 + 8 + 6 = 32 (80%); changing the second essay to 8 yields 34 (85%). Separate A-AI fixture requests 5 MC, 2 true/false, 1 identification and 2 essays, maximum 36. These fixtures must not be conflated.

8. Functional Test Cases

Test Case Format

Field	Details
Test Case ID	TC-F-014 — illustrative/unverified
Module	Reports
Test Objective	Reconcile final teacher-reviewed grades across UI, PDF and Excel.
Precondition	Continue from TC-F-013; report generation is a hypothetical failing example.
Test Input	PDF and Excel exports for A-grade and S1.
Test Steps	1. Save override. 2. Generate fresh PDF and Excel files. 3. Compare per-item score, final total, percentage and student identity with UI and database.

Field	Details
Expected Result	Both exports match the final reviewed 34/40, 85%, owner and subject filter.
Actual Result	Example: exports retained 32/40 and 80% while UI showed 34/40; BUG-001. No actual execution.
Status	ILL-FAIL; actual status NOT EXECUTED
Remarks	Hypothetical BUG-001, High priority. Tester [name - unverified]; date [not executed]; build [commit]; evidence [not captured].

Functional Testing Summary

Test Case ID	Module	Test Description	Expected Result	Illustrative Outcome	ILL Status
TC-F-001	Authentication	Sign in as each of three roles; sign out.	Correct role dashboard; protected routes require a token.	Example: all three dashboards opened; logout returned to login.	ILL-PASS
TC-F-002	Authentication	Submit wrong password and an unknown account.	Reject both attempts without creating a session or revealing account existence.	Example: generic credential errors; neither attempt created a session.	ILL-PASS
TC-F-003	Registration	Register a valid student; repeat the same email.	Create one student account; reject duplicate email; never grant admin role.	Example: one student created; duplicate refused; role remained student.	ILL-PASS

Test Case ID	Module	Test Description	Expected Result	Illustrative Outcome	ILL Status
TC-F-004	Subjects	T1 creates subject A and enrolls S1.	Persist ownership and membership; subject B owned by T2 stays inaccessible.	Example: S1 saw A; T1 could not edit B.	ILL-PASS
TC-F-005	Drafts	Save a manually authored assessment and reopen it.	Save as draft with unpublished flag; retain questions and instructions.	Example: draft persisted; student lists omitted it.	ILL-PASS
TC-F-006	Publishing	Publish a reviewed draft using the publish action.	Only explicit publication exposes the assessment to enrolled students.	Example: S1 could load the published quiz; unenrolled S2 could not.	ILL-PASS
TC-F-007	AI generation	Generate exact requested question counts from lesson text.	Return validated 5 MC, 2 true/false, 1 identification and 2 essays; save one draft.	Example: 10 valid questions; correct type counts; draft stayed unpublished.	ILL-PASS
TC-F-008	Lesson upload	Generate from equivalent PDF, DOCX, PPTX and TXT lessons.	Extract readable text in order; reject unreadable source before provider use.	Example: four readable fixtures processed; scanned-only PDF required OCR.	ILL-PASS
TC-F-009	AI validation	Return duplicate choices, missing keys or unsupported answers.	Reject invalid candidates; accept only grounded, unique, schema-valid questions.	Example: invalid candidates rejected; bounded recovery returned a valid draft.	ILL-PASS

Test Case ID	Module	Test Description	Expected Result	Illustrative Outcome	ILL Status
TC-F-010	Submission	S1 completes an open published assessment.	Persist one submission with answers, time and student/assessment references.	Example: one submitted record; teacher list displayed the same answers.	ILL-PASS
TC-F-011	Objective grading	Grade correct, incorrect and accepted-alias answers.	Award full item points only for normalized accepted answers; incorrect is zero.	Example: 9 of 10 two-point items correct; objective subtotal 18/20.	ILL-PASS
TC-F-012	Essay grading	Grade two essays and reconcile the mixed assessment total.	Whole-number rubric scores within maxima; totals equal 32/40 and 80%.	Example: essays 8/10 and 6/10; objective 18/20; total 32/40, 80%.	ILL-PASS
TC-F-013	Teacher review	Override the second essay from 6 to 8 with a reason.	Store reviewer/reason; total becomes 34/40, 85%; preserve teacher decision.	Example: review UI showed 34/40 and 85%; reason retained after refresh.	ILL-PASS
TC-F-014	Reports	Export PDF and Excel after the teacher override.	Both exports match the final reviewed 34/40, 85%, owner and subject filter.	Example: exports retained 32/40 and 80% while UI showed 34/40; BUG-001.	ILL-FAIL

Test Case ID	Module	Test Description	Expected Result	Illustrative Outcome	ILL Status
TC-F-015	Privacy	Try cross-student, cross-teacher and draft detail access.	Deny unauthorized records; authorized student payload omits keys and source text.	Example: unauthorized requests denied; student detail contained no answer keys.	ILL-PASS
TC-F-016	Notifications	Publish, submit and grade; open the notification panel.	Correct recipient and destination; read state persists; duplicate events do not multiply.	Example: S1 received assessment/grade notices; T2 received none.	ILL-PASS
TC-F-017	Quiz Arena	Join an active game, answer and view results.	Valid participation; scores and rank agree with stored attempts; no cross-subject leak.	Example: S1 joined once; leaderboard score matched the saved attempt.	ILL-PASS
TC-F-018	Lists and filters	Page through assessment summaries; filter by subject/status.	Correct 20-item page and hasMore; draft filter includes AI drafts; no lesson bulk.	Example: pages returned 20 then 5 unique summaries; filters matched.	ILL-PASS
TC-F-019	Duplicate handling	Double-click generation and retry submission.	Avoid duplicate persisted drafts and submissions for the same logical operation.	Example: one shared in-process generation; one submission record after retry.	ILL-PASS

Test Case ID	Module	Test Description	Expected Result	Illustrative Outcome	ILL Status
TC-F-020	Assessment window	Open before start, after close and after time expires.	Enforce server-authoritative availability and duration; preserve permitted answer state.	Example: early/closed starts denied; expired attempt marked timeExpired.	ILL-PASS

All functional summary outcomes are illustrative/unverified. Detailed prerequisites, inputs, steps and evidence requirements are in Appendix A. The full-format example above repeats TC-F-014 and is counted once.

9. Input Validation Testing

Test ID	Input Field	Test Condition	Expected Behavior	Illustrative Behavior	ILL Status
IV-001	Full name	Submit blank or whitespace-only name.	Reject; no account created.	Example: required-name message; no record.	ILL-PASS
IV-002	Email	Submit student@ and then duplicate .test email.	Reject malformed and duplicate email.	Example: both rejected; count unchanged.	ILL-PASS

Test ID	Input Field	Test Condition	Expected Behavior	Illustrative Behavior	ILL Status
IV-003	Password	Submit fewer than required characters; mismatch confirmation.	Enforce configured rule; show clear mismatch.	Example: rejected; no session. Exact minimum [verify].	ILL-PASS
IV-004	Essay points	Submit -1, text, and 2.5 as essay maximum.	Accept only valid positive whole-number maxima.	Example: invalid maxima rejected before save.	ILL-PASS
IV-005	Assessment title	Save blank title while bypassing client validation.	Server rejects required field; no empty assessment.	Example: validation response; no draft saved.	ILL-PASS
IV-006	Lesson file	Upload .exe, corrupt DOCX and empty TXT.	Reject unsupported/unreadable/empty data before AI.	Example: clear error; provider call count 0.	ILL-PASS
IV-007	Extracted lesson	Submit 25,001 cleaned characters.	Reject with DOCUMENT_TOO_LARGE; never truncate silently.	Example: rejected; no AI request or draft.	ILL-PASS
IV-008	Upload size	Upload exactly 10 MiB and 10 MiB + 1 byte.	Apply file limit; oversized request cannot reach AI.	Example: readable boundary accepted; larger file rejected.	ILL-PASS

Input cases IV-001 through IV-008 are independent illustrative checks. Inspect both client messaging and server rejection; bypass client validation when testing server controls. Example outcomes do not establish that validation is enforced.

10. Database Testing

10.1 Database Operations

Test ID	Operation	Test	Expected Result	Illustrative Result	ILL Status
DB-001	INSERT	Save a reviewed valid draft; count linked question records.	One owned draft; correct points and references.	Example: draft and 12 linked questions reconciled.	ILL-PASS
DB-002	READ	Load owned versus other teacher records; paginate 25 rows.	Only authorized records; stable distinct pages.	Example: 20 + 5 records; no foreign subject data.	ILL-PASS
DB-003	UPDATE	Override E2 6→8; reload submission and feedback.	Persist final 34/40, reviewer and reason.	Example: values retained; no unrelated answer changed.	ILL-PASS
DB-004	DELETE	Delete an unused teacher-owned draft; reload IDs.	Draft disappears from active views; no exposed orphan content.	Example: draft unavailable; related-data policy [verify].	ILL-PASS
DB-005	Uniqueness	Insert duplicate email and assessment/student pair.	Indexes prevent duplicate records; controlled response.	Example: both duplicate writes rejected.	ILL-PASS

Test ID	Operation	Test	Expected Result	Illustrative Result	ILL Status
DB-006	Mock recovery	Call admin backup/restore in an isolated DB; compare counts.	Explicit mock response; no claim of real backup or restore.	Example: mock message; counts unchanged; recovery unverified.	ILL-PASS

10.2 Data Integrity

Planned integrity checks; none verified:

- [Not executed] MongoDB _id uniqueness and correct record identity.
- [Not executed] Mongoose reference checks; MongoDB does not supply relational foreign keys.
- [Not executed] No orphaned subject, assessment, question, submission or feedback references.
- [Not executed] Required title, role, student and assessment references.
- [Not executed] Valid types, integer essay scores and bounded maxima.
- [Not executed] Unique email and assessment/student submission combination.
- [Not executed] Inject partial write failures and reconcile records; do not assume multi-document transactions.
- [Not executed] Admin mock response is not recovery evidence; actual restore excluded.

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11. API / External Service Testing

The API scenarios use synthetic payloads and controlled stubs. A future live run must capture provider/model, correlation ID, status, latency and redacted errors; never record keys, lesson text or student answers in shared logs.

Test ID	API/Service	Request or Fault	Expected Response	Illustrative Response	ILL Status
API-001	Assessment list	GET summary page 1, limit 20, teacher token.	items/page/limit/hasMore; no lesson/version bulk.	Example: valid 20-item summary envelope.	ILL-PASS
API-002	Student detail	GET permitted assessment, then another subject ID.	Authorized detail omits keys; unauthorized request denied.	Example: permitted data only; second request denied.	ILL-PASS
API-003	AI primary	POST generate-quiz with valid lesson and counts.	Validated exact counts; save one unpublished draft.	Example: valid Groq-stub response saved once.	ILL-PASS
API-004	AI fallback	Inject Groq 429, then valid OpenAI output.	Switch directly to fallback; validate once before persistence.	Example: one call per provider; one draft.	ILL-PASS
API-005	AI hard failure	Inject Groq 401/configuration error.	Return actionable error; do not activate fallback.	Example: fallback calls 0; saved drafts 0.	ILL-PASS

Test ID	API/Service	Request or Fault	Expected Response	Illustrative Response	ILL Status
API-006	AI validation	Inject invalid JSON, then invalid recovery output.	At most two Groq attempts, one eligible fallback; never save invalid output.	Example: bounded calls; only validated fallback draft saved.	ILL-PASS
API-007	AI timeout/cancel	Exhaust quiz budget; separately disconnect all callers.	Budget returns 504; cancellation aborts work, no later fallback/save.	Example: timeout 504; cancelled run saved 0 drafts.	ILL-PASS
API-008	Essay provider outage	Fail Groq and OpenAI while grading a nonblank essay.	Keep score/final total pending review; never invent zero.	Example: pending_review; no published final score.	ILL-PASS

API Evaluation

- [Not executed] Valid authorized requests and response schema.
- [Not executed] Invalid identifiers, missing fields and invalid page limits.
- [Not executed] Missing/expired tokens and ownership checks.
- [Not executed] Attempt timeouts and total quiz budget exhaustion.
- [Not executed] Stable error shape and actionable UI state.
- [Not executed] Schema, exact counts, uniqueness and lesson grounding.

- [Not executed] 429, Retry-After and bounded fallback.
- [Not executed] Both-provider failure, cancellation and pending teacher review.

Configured quiz defaults in project documentation are a 65,000 ms total AI budget and 20,000 ms attempt cap; actual deployment values are [verify]. An AI budget does not bound document extraction/database time. Essay grading uses a separate provider workflow; both-provider essay failure must remain pending teacher review rather than a fabricated zero.

12. Performance Testing

Test ID	Performance Metric	Test Condition	Proposed Criterion	Illustrative Measurement	ILL Status
PT-001	Dashboard load	1 user; 30 loads, after warm-up.	Proposed p95 ≤3.0 s.	Example: p95 1.90 s; not measured.	ILL-PASS
PT-002	Summary API	1 user; 200 completed requests, same 20-row payload.	Proposed p95 ≤1.5 s; errors ≤1%.	Example: mean .62 s; p95 1.20 s; 0/200 errors.	ILL-PASS
PT-003	Database read	200 indexed ownership-filtered reads; seeded 10,000 submissions.	Proposed p95 ≤200 ms; capture explain plan.	Example: p95 85 ms; not measured.	ILL-PASS
PT-004	Concurrent load	50 virtual users; 10 min; 2,000 requests; AI stubbed.	Proposed p95 ≤3.0 s; errors ≤1%; no lost grades.	Example: p95 2.40 s; 4/2,000 errors; no loss.	ILL-PASS
PT-005	Stress load	100 virtual users; 10 min; 2,000 requests; same mix.	Proposed p95 ≤5.0 s; errors ≤2%; no lost grades.	Example: p95 7.80 s; 64/2,000 errors; BUG-002.	ILL-FAIL

Performance Metrics

- Illustrative average response time: 0.62 s for PT-002 only; unverified.
- Illustrative minimum response time: 0.21 s for the same PT-002 sample; unverified.
- Illustrative maximum response time: 1.84 s for the same PT-002 sample; unverified.
- Assumed concurrent users: 1 for PT-002; 50 for PT-004; 100 for PT-005.
- Illustrative error rate: PT-002 0/200 = 0%; PT-004 4/2,000 = 0.20%; PT-005 64/2,000 = 3.20%. No measurements taken.
- Actual system availability: [Not measured]. A short synthetic load example cannot establish uptime or an SLA.

Proposed protocol: record [build], [UTC/Asia-Manila timestamps], test runner, host resources and network conditions. Warm up 2 minutes, then run a 10-minute sample per workload. Capture client timings and server/query telemetry separately. Use 20% login/session, 40% list/detail, 20% dashboard and 20% submission requests; stub AI for load isolation, then assess live AI separately. Numerical values below are invented examples.

13. Security Testing

Test ID	Security Area	Test Procedure	Expected Result	Illustrative Result	ILL Status
SEC-001	Authentication	Wrong password; missing and expired token.	Deny access; no token/session issued for failure.	Example: protected data withheld.	ILL-PASS
SEC-002	Authorization	Use student token on admin API; use T1 token on T2 records.	Enforce role and ownership server-side.	Example: both requests denied.	ILL-PASS
SEC-003	Passwords	Inspect synthetic user storage and API responses.	Hash with bcrypt; never expose password/hash to client.	Example: hash stored; credential fields omitted.	ILL-PASS

Test ID	Security Area	Test Procedure	Expected Result	Illustrative Result	ILL Status
SEC-004	Client session	Logout, Back, refresh; request API without stored token.	Clear local session; protected request denied.	Example: login required. Stolen-token revocation not established.	ILL-PASS
SEC-005	Unsafe input	Submit script-like text and object/operator login fields.	No script execution or auth bypass; reject invalid types.	Example: text inert; malformed credentials rejected.	ILL-PASS
SEC-006	Assessment privacy	Inspect student detail payload and another student submission URL.	No answer keys/source/rubric answers; no cross-student data.	Example: keys absent; foreign submission denied.	ILL-PASS

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14. Compatibility Testing

Browser Compatibility

Browser / ID	Version	ILL Result	Illustrative Remarks
BR-001 Chrome	[TBD]	ILL-PASS	Assumed Windows 11. Example: login, quiz, review, exports usable.
BR-002 Edge	[TBD]	ILL-PASS	Assumed Windows 11. Example: same workflow completed.

Browser / ID	Version	ILL Result	Illustrative Remarks
BR-003 Firefox	[TBD]	ILL-PASS	Assumed Windows 11. Example: no blocking interaction failure.
BR-004 Safari	[TBD]	ILL-PASS	Assumed macOS [release]. Example: quiz and download workflow usable.

Device Compatibility

Device / ID	Assumed OS / Browser	Assumed Screen Size	Illustrative Result
DEV-001 Desktop	Assumed Windows 11 / Chrome	Assumed 1920×1080	ILL-PASS; example: workflow usable.
DEV-002 Laptop	Assumed Windows 11 / Edge	Assumed 1366×768	ILL-PASS; example: controls reachable.
DEV-003 Tablet	Assumed iPadOS [release] / Safari	Assumed 820×1180 CSS px	ILL-PASS; example: touch workflow usable.
DEV-004 Smartphone	Assumed Android [release] / Chrome	Assumed 390×844 CSS px	ILL-FAIL; example: review Save clipped; BUG-003.

Compatibility IDs: BR-001..004 and DEV-001..004; 8 separate illustrative cases. Browser versions and exact OS releases are placeholders. Repeat login, quiz completion, review and export on each listed combination; record version and device evidence during actual execution.

15. Usability and User Interface Testing

Test ID	Usability Criterion	Expected Result	Illustrative Result	ILL Status
UI-001	Navigation	Reach assigned assessment in ≤3 intentional selections.	Example: student reached quiz in 2 selections.	ILL-PASS
UI-002	Readability	Labels and grade statuses legible at 200% zoom.	Example: key labels legible; focus remained visible.	ILL-PASS
UI-003	Consistency	Same pending/reviewed grade meaning across list/detail.	Example: pending label consistent; no false zero.	ILL-PASS
UI-004	Feedback	Show generation/submission pending; prevent accidental duplicate action.	Example: progress indicator and disabled submit shown.	ILL-PASS

Test ID	Usability Criterion	Expected Result	Illustrative Result	ILL Status
UI-005	Error messages	Provider/upload failure offers actionable cause and preserves input.	Example: file error clear; lesson input retained.	ILL-PASS
UI-006	Responsive review	Teacher reaches Save by touch/keyboard at 390 px width.	Example: Save clipped off-screen; BUG-003.	ILL-FAIL

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16. ISO/IEC 25010 Software Quality Evaluation

Use ISO/IEC 25010:2023 as an organizing framework for the nine quality characteristics already listed in this template. This is an illustrative coverage mapping, not a conformance assessment, certification or expert rating. N/E means not evaluated; all nine actual quality evaluations are pending.

Quality Characteristic	Evaluation Focus	Result
Functional Suitability	Assessment/grading correctness; F-001..020 and IV checks. Illustrative gap: BUG-001; actual evaluation pending.	N/E
Performance Efficiency	PT-001..005; latency, errors and host-resource capture. Illustrative gap: BUG-002; resources unmeasured.	N/E
Compatibility	BR/DEV matrices and interoperable PDF/Excel outputs. Illustrative mobile gap: BUG-003; versions unverified.	N/E

Quality Characteristic	Evaluation Focus	Result
Interaction Capability	UI-001..006; navigation, clarity and accessible controls. Illustrative gap: UI-006; user evaluation pending.	N/E
Reliability	DB/API fault handling; pending review; recovery plan. Unverified; real backup/restore excluded.	N/E
Security	SEC-001..006; ownership and answer confidentiality. Unverified; examples are not penetration evidence.	N/E
Maintainability	Existing modular code/test structure; future repeatable fixtures. Not rated; suite not run for this report.	N/E
Flexibility	Environment setup, model configuration and scaled deployment. Not rated; install/scaling not demonstrated.	N/E
Safety	Prevent false grades; human review and correction trail. Not rated; grade-impact risk requires verified review controls.	N/E

Edition reference: ISO/IEC 25010:2023, second edition, product quality model; https://www.iso.org/standard/78176.html. No numerical quality score is assigned without defined measures and actual observations. ISO mapping rows are excluded from test-case totals.

17. Defect / Bug Report

Defect ID	Module	Description	Severity	Priority	Status
BUG-001	Grade reports	Hypothetical: final UI 34/40 (85%); fresh PDF/Excel 32/40 (80%); F-014.	High (example)	High (example)	Hypothetical open; unverified
BUG-002	Capacity	Hypothetical: PT-005 p95 7.80 s; 3.20% errors exceeds proposed stress limits.	Medium (example)	Medium (example)	Hypothetical open; unverified
BUG-003	Mobile review	Hypothetical: Save unreachable at 390 px; DEV-004 and UI-006.	Medium (example)	Medium (example)	Hypothetical open; unverified

Severity Classification

Severity	Description
Critical	Prevents the system from operating or causes major data/security failure
High	Major function does not work correctly
Medium	Function works with limitations or workaround
Low	Minor defect with limited impact

Every BUG entry below is hypothetical and unverified; none is a confirmed issue in the current build. Severity describes the example impact only. No actual ticket, fix, developer identity or closure is claimed. Appendix E contains reproducible hypothetical records.

18. Corrective Action and Retesting

Defect ID	Corrective Action	Date Fixed	Retest Result	Final Status
BUG-001	Proposed: reconcile all exports to final reviewed scores; add discrepancy check.	[Not fixed / date unknown]	NOT EXECUTED; repeat F-013/014 and REG-005.	Unverified example; no closure
BUG-002	Proposed: profile slow requests/indexes and bound expensive work; rerun same load.	[Not fixed / date unknown]	NOT EXECUTED; repeat PT-002/004/005.	Unverified example; no closure
BUG-003	Proposed: correct responsive container/button reachability and keyboard focus.	[Not fixed / date unknown]	NOT EXECUTED; repeat DEV-004 and UI-006.	Unverified example; no closure

Corrective actions are proposals only. Date-fixed fields remain placeholders; retest state is NOT EXECUTED. A defect may be closed only after the referenced case passes with retained evidence on a named build.

19. Regression Testing

REG-001..005 show illustrative baseline regression outcomes only. They are not post-fix evidence and do not close BUG-001..003. After any actual correction, rerun these checks plus the affected failing cases, using the same fixture and recorded build.

Test ID	Baseline Function	Illustrative Outcome	ILL Status
REG-001	Login/logout and role routing	Example: three roles routed correctly; no post-fix run.	ILL-PASS
REG-002	Registration and duplicate prevention	Example: valid account created once; no post-fix run.	ILL-PASS
REG-003	Draft persistence and page filters	Example: save/reload/page IDs consistent; no post-fix run.	ILL-PASS
REG-004	AI drafts and objective/essay totals	Example: unpublished draft and 32/40 totals; no post-fix run.	ILL-PASS

Test ID	Baseline Function	Illustrative Outcome	ILL Status
REG-005	Baseline reports before override	Example: both exports show baseline 32/40; does not resolve F-014.	ILL-PASS

---

20. Testing Results Summary

Example Category	Example Cases	ILL Passed	ILL Failed	ILL Pass Rate
Functional	20	19	1	95.00%
Input Validation	8	8	0	100.00%
Database	6	6	0	100.00%
API	8	8	0	100.00%
Performance	5	4	1	80.00%
Security	6	6	0	100.00%

Example Category	Example Cases	ILL Passed	ILL Failed	ILL Pass Rate
Compatibility	8	7	1	87.50%
Usability	6	5	1	83.33%
Regression	5	5	0	100.00%
TOTAL (illustrative)	72	68	4	94.44%

Pass Rate Formula

Illustrative pass rate = (68 illustrative passes / 72 example cases) × 100 = 94.44%. Actual executions = 0; actual pass rate = not available.

The table counts one example outcome per unique ID: F 20, IV 8, DB 6, API 8, PT 5, SEC 6, BR/DEV 8, UI 6 and REG 5. Four failing cases map to three hypothetical defects: F-014 → BUG-001; PT-005 → BUG-002; DEV-004 and UI-006 → BUG-003. Repeated details in Appendices A/E and the full-format case are not additional tests.

21. Overall Technical Assessment

Overall Testing Result

☐ Passed – Ready for Deployment [not assessed]

☐ Passed with Minor Issues [not assessed]

☐ For Further Improvement [actual evaluation pending]

☐ Failed – Major Corrections Required [not assessed]

Technical Assessment

Based on the illustrative scenario set, the proposed verification covers:

> The assessment lifecycle, score arithmetic, AI validation/fallback, teacher review, ownership, reporting and interface behavior. Hypothetical results show why report reconciliation, capacity and mobile access require explicit checks. No actual functionality, security, AI accuracy or readiness conclusion can be drawn from this report.

The hypothetical defects and issues are:

> One high-severity grade-export discrepancy and two medium-severity capacity/mobile issues. All remain unverified example records with retesting pending. The sample 94.44% pass rate is a document example only.

Actual technical assessment:

☐ Technically Acceptable ☐ Acceptable with Minor Revisions ☐ Requires Further Testing ☐ Not Acceptable

---

No assessment option is selected. Actual testing, independent review and signed approval are pending. Deployment authorization is not granted by this report.

22. Recommendations

The following actions are proposed for a future verified testing cycle:

1. Freeze [build/commit] and obtain the institution-approved scoring, publication, retention and access requirements before executing the 72 scenarios.

2. Reproduce any suspected report discrepancy; reconcile final reviewed scores across UI, database, PDF and Excel; retain the teacher review reason and audit evidence.
3. Measure realistic concurrency with controlled workloads, payload sizes and AI latency; use actual percentiles and error counts before setting capacity claims.
4. Verify mobile access, keyboard navigation and clear pending/error states; use human review for consequential AI scoring and treat integrity indicators as review cues.
5. Implement and rehearse real database backup/restore separately from admin mocks; verify privacy controls and provider configuration in the intended deployment.

Obtain actual programmer/analyst and 10-IT-expert participation required by the template; attach identities, execution records, corrections, retests and approval signatures only when verified.

23. Testing Approval

Role	Name	Signature	Date
Test Engineer/Tester	[Name - unverified]	[Unsigned]	[Approval date pending]
Developer	[Name - unverified]	[Unsigned]	[Approval date pending]
Project Leader	[Name - unverified]	[Unsigned]	[Approval date pending]
Technical Adviser	[Name - unverified]	[Unsigned]	[Approval date pending]
Project Adviser	[Name - unverified]	[Unsigned]	[Approval date pending]

Approval is pending. All names, signatures and dates below are unknown placeholders; no tester or evaluator identity has been supplied or verified. This illustrative draft must not be represented as an executed or signed report.

24. Appendices

Appendix A – Test Case Documentation

Detailed functional scenarios follow. Each case uses the same hypothetical outcome and ILL status as Section 8. Tester: [Tester name - unverified]; execution date/time: [Not executed]; build: [Build/commit]. Planned evidence is a capture requirement, not an existing attachment.

TC-F-001 Authentication

Precondition: Active synthetic admin, teacher T1 and student S1; clean browser session. Input: Role accounts from the synthetic fixture; [valid test passwords].

Steps: 1. Sign in as each role in a separate session. 2. Open the dashboard and a protected URL. 3. Sign out and use Back, then refresh.

Expected: Correct role dashboard; protected routes require a token. Illustrative/unverified outcome: Example: all three dashboards opened; logout returned to login. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-002 Authentication

Precondition: T1 exists; unknown@example.test does not. Input: T1 with a wrong password; unknown@example.test with the same password.

Steps: 1. Submit each credential pair. 2. Inspect returned message and session storage. 3. Request a protected endpoint without a token.

Expected: Reject both attempts without creating a session or revealing account existence. Illustrative/unverified outcome: Example: generic credential errors; neither attempt created a session. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-003 Registration

Precondition: Disposable database; registration permitted; no account with new.student@example.test. Input: Synthetic name; unique .test email; valid password; required agreements.

Steps: 1. Complete student registration. 2. Sign in. 3. Repeat the same email. 4. Inspect role and account count.

Expected: Create one student account; reject duplicate email; never grant admin role. Illustrative/unverified outcome: Example: one student created; duplicate refused; role remained student. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-004 Subjects

Precondition: T1 and T2 exist; S1 is not enrolled; subjects A and B have separate owners. Input: Subject A; S1 synthetic user identifier.

Steps: 1. Create A as T1. 2. Add S1. 3. Reload as S1. 4. Attempt to edit B as T1.

Expected: Persist ownership and membership; subject B owned by T2 stays inaccessible. Illustrative/unverified outcome: Example: S1 saw A; T1 could not edit B. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-005 Drafts

Precondition: T1 owns A; S1 is enrolled in A. Input: A-draft; two MC questions at 2 points each; instructions.

Steps: 1. Create assessment. 2. Add questions and save. 3. Reopen as T1. 4. Check student list and direct access as S1.

Expected: Save as draft with unpublished flag; retain questions and instructions. Illustrative/unverified outcome: Example: draft persisted; student lists omitted it. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-006 Publishing

Precondition: A-draft is valid and reviewed; S1 enrolled, S2 unenrolled; window is open. Input: A-draft identifier; valid start/end and duration.

Steps: 1. Confirm draft is hidden. 2. Publish as T1. 3. Reload as S1 and S2. 4. Inspect publication state.

Expected: Only explicit publication exposes the assessment to enrolled students. Illustrative/unverified outcome: Example: S1 could load the published quiz; unenrolled S2 could not. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-007 AI generation

Precondition: T1 owns A; provider transport is stubbed or isolated test credentials are configured. Input: Synthetic electricity lesson; counts 5/2/1/2; MC/TF/ID worth 2, essays 10.

Steps: 1. Supply lesson and counts. 2. Generate once. 3. Inspect type counts, keys, options and source grounding. 4. Check saved draft and student visibility.

Expected: Return validated 5 MC, 2 true/false, 1 identification and 2 essays; save one draft. Illustrative/unverified outcome: Example: 10 valid questions; correct type counts; draft stayed unpublished. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-008 Lesson upload

Precondition: Four synthetic lessons contain the same concepts; no real student data. Input: Four files below 10 MiB and 25,000 extracted characters; scanned PDF control.

Steps: 1. Upload each readable file separately. 2. Compare extracted concepts. 3. Generate and review. 4. Upload scanned-only control and inspect error and provider call count.

Expected: Extract readable text in order; reject unreadable source before provider use. Illustrative/unverified outcome: Example: four readable fixtures processed; scanned-only PDF required OCR. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-009 AI validation

Precondition: Provider stub can return malformed candidates followed by valid recovery output. Input: Candidate with duplicate MC options; candidate missing answerKey; off-topic key.

Steps: 1. Inject each invalid response. 2. Run generation. 3. Inspect validation and recovery limits. 4. Confirm invalid questions were never persisted.

Expected: Reject invalid candidates; accept only grounded, unique, schema-valid questions. Illustrative/unverified outcome: Example: invalid candidates rejected; bounded recovery returned a valid draft. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-010 Submission

Precondition: S1 enrolled in A; A-grade published and open; no existing S1 submission. Input: Complete A-grade answers; agreed synthetic start/end times.

Steps: 1. Start assessment. 2. Answer each item. 3. Submit once. 4. Reload student history and teacher submissions.

Expected: Persist one submission with answers, time and student/assessment references. Illustrative/unverified outcome: Example: one submitted record; teacher list displayed the same answers. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-011 Objective grading

Precondition: Separate objective fixture with 10 items, each worth 2; fixed reviewed answer keys. Input: Nine correct answers, including one allowed alias, and one wrong answer.

Steps: 1. Submit fixture. 2. Compare each answer to teacher keys. 3. Sum item scores. 4. Confirm subtotal and rounding.

Expected: Award full item points only for normalized accepted answers; incorrect is zero. Illustrative/unverified outcome: Example: 9 of 10 two-point items correct; objective subtotal 18/20. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-012 Essay grading

Precondition: A-grade: 10 objective items at 2 and two essays at 10; reviewed rubric and lesson. Input: Objective subtotal 18; essay responses E1/E2; valid stub scores 8 and 6.

Steps: 1. Submit mixed assessment. 2. Wait for processing to finish. 3. Review criterion sums and bounds. 4. Check teacher, student and stored totals.

Expected: Whole-number rubric scores within maxima; totals equal 32/40 and 80%. Illustrative/unverified outcome: Example: essays 8/10 and 6/10; objective 18/20; total 32/40, 80%. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-013 Teacher review

Precondition: A-grade has the TC-F-012 result; T1 owns A; pending AI task can be delayed. Input: Second essay final score 8; reason: credit for correct supporting explanation.

Steps: 1. Open answer review as T1. 2. Set score 8 and reason. 3. Save and reload. 4. Release delayed AI result and confirm it cannot overwrite review.

Expected: Store reviewer/reason; total becomes 34/40, 85%; preserve teacher decision. Illustrative/unverified outcome: Example: review UI showed 34/40 and 85%; reason retained after refresh. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-014 Reports

Precondition: Continue from TC-F-013; report generation is a hypothetical failing example. Input: PDF and Excel exports for A-grade and S1.

Steps: 1. Save override. 2. Generate fresh PDF and Excel files. 3. Compare per-item score, final total, percentage and student identity with UI and database.

Expected: Both exports match the final reviewed 34/40, 85%, owner and subject filter. Illustrative/unverified outcome: Example: exports retained 32/40 and 80% while UI showed 34/40; BUG-001. Status: ILL-FAIL. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-015 Privacy

Precondition: T1/T2 own separate subjects; S1/S2 have separate submissions. Input: Another user submission ID; draft ID; published authorized ID.

Steps: 1. Exchange IDs in direct requests under each role. 2. Inspect status and payload. 3. Load a permitted student assessment and search for keys, rubric answers and lessonContent.

Expected: Deny unauthorized records; authorized student payload omits keys and source text. Illustrative/unverified outcome: Example: unauthorized requests denied; student detail contained no answer keys. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-016 Notifications

Precondition: S1 enrolled in A; T2 has no connection to A; fresh notification fixture. Input: Publication, submission and grading events; repeated same event identifier.

Steps: 1. Trigger each event. 2. Open S1 panel. 3. Follow destination and mark read. 4. Reload and replay the same event.

Expected: Correct recipient and destination; read state persists; duplicate events do not multiply. Illustrative/unverified outcome: Example: S1 received assessment/grade notices; T2 received none. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-017 Quiz Arena

Precondition: T1 hosts a synthetic game for A; S1 eligible; S2 unauthorized. Input: Valid game code; one correct and one incorrect response; invalid-code control.

Steps: 1. Join as S1. 2. Submit responses. 3. Finish and compare results. 4. Retry invalid code and unauthorized S2 access.

Expected: Valid participation; scores and rank agree with stored attempts; no cross-subject leak. Illustrative/unverified outcome: Example: S1 joined once; leaderboard score matched the saved attempt. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-018 Lists and filters

Precondition: A list fixture has exactly 25 teacher-owned assessments and controlled statuses. Input: view=summary; page 1 then 2; limit 20; subject A; draft status.

Steps: 1. Load page 1 and 2. 2. Compare IDs for omissions/duplicates. 3. Apply filters. 4. Inspect summary payload for unnecessary lesson/version content.

Expected: Correct 20-item page and hasMore; draft filter includes AI drafts; no lesson bulk. Illustrative/unverified outcome: Example: pages returned 20 then 5 unique summaries; filters matched. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-019 Duplicate handling

Precondition: Single backend instance; identical teacher/content requests; submission unique index present. Input: Two concurrent identical generation requests; same S1 assessment submitted twice.

Steps: 1. Send both generation requests concurrently. 2. Count provider calls and saved drafts. 3. Retry submission. 4. Inspect database count. Cross-instance behavior requires separate validation.

Expected: Avoid duplicate persisted drafts and submissions for the same logical operation. Illustrative/unverified outcome: Example: one shared in-process generation; one submission record after retry. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

TC-F-020 Assessment window

Precondition: Disposable assessment sessions; agreed timezone; controlled clock offsets. Input: Start minus 1 second; end plus 1 second; duration limit plus 1 second.

Steps: 1. Attempt start at each boundary. 2. Run one open attempt past duration. 3. Refresh timer and submit. 4. Inspect availability response and persisted expiry flag.

Expected: Enforce server-authoritative availability and duration; preserve permitted answer state. Illustrative/unverified outcome: Example: early/closed starts denied; expired attempt marked timeExpired. Status: ILL-PASS. Actual: NOT EXECUTED. Evidence to capture: redacted UI/API response and relevant record/export comparison, linked to this ID.

Appendix B – Screenshots of Testing

Actual screenshots: [Not captured - no tests executed]. Evidence register for a future run:

- [Screenshot F-006] Teacher publish state and enrolled student view; redact personal data.
- [Screenshot F-014] Compare final reviewed grade with both exports if a discrepancy is reproduced.
- [Screenshot API-008] Provider-failure message and pending-review state; omit keys and answers.
- [Screenshot BUG-001..003] Before/after capture on named builds after a verified correction.
- [Screenshot retest] Repeated failing case, actual status, tester and timestamp after correction.

Appendix C – System Logs

Actual logs: [Not collected]. Required future evidence, stored with restricted access:

- [Application log] Correlation ID, route, role and result; no raw student responses.
- [API log] Provider/model, failure class, retry/fallback, elapsed time and request ID; no API secrets or lesson text.
- [Error log] Reproduced failure, stack and sanitized context linked to a verified defect ID.
- [Database evidence] Record IDs/counts, score reconciliation and index presence; no production dump.
- [Performance evidence] Raw timestamps, request/error counts, percentile method, resources and workload settings.

Appendix D – Test Environment

Assumed topology only: one React browser client → Express/Node API → isolated MongoDB; outbound Groq and OpenAI connections where enabled. One backend instance for process-local deduplication. Frontend URL [Test URL]; API URL [Test API URL]; database [Isolated test DB]; commit [Build/commit]; run window [Date/time Asia/Manila]; network trace [Evidence path]. No deployment was inspected.

Environment verification record: CPU/RAM/storage [verify]; OS [version]; browser [version]; Node/React/Express/MongoDB [exact versions]; dependency lockfile [hash]; indexes [verified list]; TLS/proxy timeout [value]; provider/model IDs [values]; test runner [tool/version]. These are placeholders, not observations.

Assumed fault setup: stub provider transport for 429, 401, malformed JSON, timeout and both-provider failure; use controlled clock values for availability checks. Production secrets must not be copied into the report. Record only sanitized configuration and request correlation IDs.

Source context: local README.md and relevant routes/models/services reviewed as static project material, including Submission, grading, submission review, upload policy and admin mock recovery handlers. Existing test files or prior diagnostic files are not treated as current execution evidence. No application code or database was changed for this report.

Appendix E – Defect Documentation

All records below are hypothetical/unverified defect documentation. Owners, dates, evidence and fixes remain placeholders. They are examples of reporting practice, not findings from executed tests.

BUG-001 Hypothetical grade report discrepancy

Status: hypothetical open/unverified. Reporter [Tester name - unverified]; owner [Developer name]; found date [Not executed]; build [Build/commit]; evidence [Not captured]. Severity High and priority High are illustrative only.

Reproduction proposal: use A-grade with objective subtotal 18 and essays 8/6. Confirm baseline 32/40 (80%). Change E2 to 8 with a reason; confirm UI 34/40 (85%). Generate new PDF and Excel exports. Expected: both 34/40 (85%). Hypothetical observed result: both still 32/40 (80%). Impact: a five-percentage-point reporting discrepancy.

Root cause: unknown/unverified; a stale aggregate or use of an AI score rather than final reviewed score is only a hypothesis. Proposed correction: reconcile the report data path and cache invalidation if relevant. Retest F-013, F-014 and REG-005 on [fixed build], including a late AI completion. No fix or retest performed.

BUG-002 Hypothetical stress capacity issue

Status: hypothetical open/unverified. Reporter [Tester name - unverified]; owner [Developer name]; date [Not executed]; build [Build/commit]; raw load evidence [Not collected]. Severity Medium and priority Medium are illustrative only.

Reproduction proposal: single API instance, assumed 2 vCPU/4 GB RAM, isolated DB, seeded 10,000 submissions; use the Section 12 request mix and stub AI. Warm up 2 minutes; run 100 virtual users for 10 minutes. Proposed limits: p95 ≤5.0 s, errors ≤2%. Hypothetical result: p95 7.80 s, 64 errors in 2,000 requests (3.20%). No actual request or timing data exists.

Root cause: unknown/unverified; inspect traces and indexes before assigning a cause. Proposed correction: optimize demonstrated bottlenecks and bound expensive work. Retest PT-002, PT-004 and PT-005 with the same data/mix and recorded resources. No optimization or retest performed.

BUG-003 Hypothetical mobile review control issue

Status: hypothetical open/unverified. Reporter [Tester name - unverified]; owner [Developer name]; date [Not executed]; browser/OS [Exact versions]; build [Build/commit]; screenshot [Not captured]. Severity Medium and priority Medium are illustrative only.

Reproduction proposal: sign in as T1 on an assumed 390×844 CSS-pixel Android/Chrome viewport; open A-grade answer review; scroll and navigate focus to Save. Expected: Save reachable and activatable by touch and keyboard. Hypothetical result: control clipped outside the visible review container. Same issue is counted in DEV-004 and UI-006.

Root cause: unknown/unverified; overflow sizing is a hypothesis. Proposed correction: inspect container width/overflow and button/focus layout. Retest DEV-004/UI-006 at 390 px, then desktop/tablet regression; attach real screenshots and focus evidence. No UI change or retest performed.